

# Francisco Cruzado

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## PROFESSIONAL EXPERIENCE

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### Software Engineer | BlueScope

*June 2023 - May 2024 | Kansas City, MO*

- Co-developed a lightweight C11 server deployed at 7 US steel plants to standardize production tracking; replaced ad hoc legacy methods (e.g., Excel uploads, manual estimates) with automated SCADA data capture via PLC-connected industrial cameras.
- Integrated the plant-side server with an Azure backend over a secure VPN, using gRPC to stream real-time production data that updated Oracle 11i tables and Power BI dashboards — replacing weekly refreshes with live monitoring for critical KPIs.
- Optimized PL/SQL queries and conceived a caching strategy using Redis, reducing Oracle APEX Forms' load times by 25% on average.
- Reduced the financial reporting calculation duration by 25% by designing new database tables for frequently accessed intermediate data, reducing query loading.
- Engineered Python solution to simplify PL/SQL queries, resulting in data modules accessible through Power BI; reduced model refresh time by 33%.

### Software Engineer | United Nations Volunteers

*Jan 2023 - June 2023 | Remote*

- Updated over 40 database triggers on SQL Server, reducing refresh times across Power BI workspaces by 2 minutes.
- Co-developed a Java-based RESTful API to manage secure access to applicant documents on the UN Volunteers platform; enforced role-based controls and logging, preventing over 15 unauthorized access attempts within the first 2 months while supporting 2000+ monthly applications.
- Enabled the rollout of a file-explorer-style UI using React v16 by structuring API responses to mirror an intuitive folder hierarchy; adoption reached 85% within two months.
- Proposed and deployed a lightweight Go server on an Azure VM to replace a Node.js backend supporting UNDP's KPI dashboards; reduced complex SQL Server query response times by over 55%.
- Rewrote rendering logic in JavaScript for compatibility with existing D3.js visualizations, cutting total dashboard load times by up to 75%, improving accessibility for UNDP managers worldwide.
- Authored React and VueJS UI components to be used by 10 UN specialized agencies in WordPress.

### Software Engineer Intern | United Nations

*June 2022 - Dec 2022 | Remote*

- Optimized server requests for a trade simulation application by writing over 25 PL/SQL stored procedures.
- Developed a local document processing pipeline using Elasticsearch analyzers (Pattern, Stop, and Whitespace) to convert unstructured PDF and Word files of free trade agreements into structured XML format, enabling downstream parsing and classification.
- Deployed the pipeline to a lightweight Azure VM, automating the batch conversion of over 550 database file blobs into LLM-parsable XML hierarchies.
- Engineered 5 Elasticsearch ingest pipelines to extract and index treaty subsections — including articles, appendices, tables, and outcome projections — enabling section-level access instead of full-document querying.
- Reduced query latency by 0.5 seconds and improved search relevance, allowing users to perform targeted semantic searches across thousands of legal clauses, appendices, and quantifiable treaty metrics.

### Database Developer | Green Vex SAC

*Jan 2022 - June 2022 | Remote*

- Centralized data management by translating over 400 Excel files into a MySQL sales and inventory database.
- Secured customer and field-sensitive agricultural data by integrating Laravel-based two-factor authentication into the phpMyAdmin interface; the system blocked over 25 targeted database attacks within 3 months post-deployment.
- Configured a Redis instance to manage records for over 500 unique daily customer logins.

## PROJECTS

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### H1B Sponsor Checker | Chrome Web Store

*Dec 2022 | 1000+ users*

- Set up 3 persistent ETL pipelines for managing user donations, telemetry, and classification data using AWS RDS, SQS, and DynamoDB.
- Published a browser extension using TypeScript and Spring Boot to classify job postings by H-1B sponsorship

likelihood, leveraging a local SQLite3 database of known sponsors and keyword-based JS classifiers to display live visual cues in the DOM.

- Designed a server-side queue system to process new job postings using a LIBSVM-trained classifier hosted on a GCP VM, expanding the sponsorship database and improving classification accuracy with each user interaction.

#### **grabbiel.com | Website**

*Dec 2024 | Entertainment*

- Designed and deployed a modular fullstack architecture on a single GCP VM, powering a content publishing website with support for blogs, reels, games, and forums.
- Developed HTMX-based frontend components (search, swipe navigation, gesture handling) alongside C++ backend services for publishing, media processing, email handling, telemetry, and secure forum access.
- Built CI/CD deployment via GitHub Actions for each service, supporting content ingestion pipelines, SQLite3-based metadata storage, and media resolution workflows integrated with Google Cloud Storage.

## **EDUCATION**

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### **MEng Computer Science | Oregon State University**

*Sep 2024 - June 2026 | Corvallis, OR*

Courses: Computer Networks, Computer Architecture, Graphics Shaders, Parallel Programming

GPA: 4.0

### **BA Economics | St. Mary's University**

*Aug 2019 - May 2023 | San Antonio, TX*

GPA: 3.82

## **TECHNICAL SKILLS AND LANGUAGES**

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**Technologies:** Azure (Virtual Machines), Blender, CSS, EC2, GCC, Git, GitLab, GNU Debugger, GCP, HTML, Linux, MySQL, Oracle APEX, Oracle EBS, Postman, Power BI, PVCS, PL/SQL, RESTful APIs, Tmux, Vim

**Programming Languages:** C++, Python, Java, JavaScript, TypeScript, Golang, x86 assembly

**Languages:** Spanish (Native), English (Proficient), French (Proficient)